

MATAR LUDOVIC Country of residence: FRANCE Nationality: FRENCH linkedin.com/in/ludovicmatar	Bordeaux +33 7 81 16 64 87 Ludovic.matar@yahoo.com https://ludoviccccc.github.io/
Summary	
Research and Development Machine Learning Engineer at the French National Institute for Research in Digital Science and Technology. Specialized in statistics, probability and computer science. Graduated from a MSc in applied mathematics and a MSC in fluid mechanics.	
Work Experiences	
Research and Development Machine Learning Engineer at the French National Institute for Research in Digital Science and Technology (INRIA) - Flowers AI & CogSci Lab. From the 1st of February 2025 to the 30th of November 2026. Inria Center at the University of Bordeaux, France - I develop machine learning models to identify sources of interference occurring in simulated multi-core embedded architectures, using automated discovery methodology, that is, curiosity based algorithms. - Implemented algorithms produce databases with greater diversity measures than other methods. Ongoing development of a semantic behaviour space better characterizing interference using BERT. - Monitoring a 6 months end-of-study internship. From the 1st of April to the 30th of September 2026.	
Research Internship at French Alternative Energies and Atomic Energy Commission (CEA) - Laboratory of Artificial Intelligence and Data Science (LIAD, SGLS,DM2S, ISAS, DES). From March to August 2024. Saclay, Île-de-France, France - I developed machine learning models to characterize inherent mechanisms of heat exchange until the boiling crisis. - Works provided research tools for the Laboratory of Modelling and Simulation in Fluid Mechanics (CEA). - Deep learning methods (computer vision) were used on infrared videos to extract latent space systems of differential equations capturing relevant dynamics by symbolic regression approaches. I used architectures such as CNNs, RNNs combined with autoencoders. - Python · Deep Learning · Computer vision · Statistics · Fluid mechanics	
Research and Development Internship at Renault Group. From May to September 2023. Guyancourt, Île-de-France, France - I streamlined the creation of model order reduction techniques to accelerate the evaluation of external aerodynamic performances, making the execution of some computational fluid dynamics analysis optional. - Obtained models predict vehicles' drag coefficients with an accuracy higher than 80% using morphing surfaces, numerical simulation results and geometric data. - Python · Deep Learning · Statistics · Geometry · Fluid mechanics	
Tutoring high school students and undergraduates in mathematics since 2023. Paris, Île-de-France, France	
Special Skills	
Languages : French: native, English: C1 (5.5 at toefl test, 8th of April 2026) Control of development environments : Solid linux shell + vim. Solid git skills. Python : Proficient with pytorch, pandas, scikit-learn, numpy, matplotlib Statistics: Parametric statistics, Hypothesis test, Bayesian formulation, Time series analysis, Non parametric statistics Machine Learning, Data Science : Good knowledge of supervised and unsupervised learning, Deep Learning, (Deep) Reinforcement learning, Generative deep learning, Natural language processing, LLM skills.	
Education	
2022 – 2024 Faculté des sciences et ingénierie de Sorbonne Université, Paris First and second year of master’s Degree – Modeling and Simulation in Hydrodynamics (M1 MF2A & M2 MSH, with honors.) First year: 62.2/100. Second Year : 68.4/100.	
2021 – 2022 Université Paris Cité, Paris Second year of master’s Degree – Random Modeling Data Science (M2 MO, applied mathematics degree. 57.7/100)	
2020 – 2021 Université Paris 1 Panthéon – Sorbonne, Paris First year of master’s Degree – Mathematics Applied to Economics & Finance (M1 MAEF, with honors. 64.1/100)	
2017 – 2020 Université Paris-Saclay, Orsay Bachelor’s Degree - Mathematics (L3 MINT, with honors. 60.0/100)	
Projects I enjoyed	
Personal projects: ← link (github) . My GitHub Ludoviccccc shows some selected machine learning and fluid mechanics projects which I've worked on. - Implementation of Policy gradient, actor critic and double Q learning with a home made grid-world environment. - Double Q-learning with replay buffer with some open AI gym environments.	
Projects: Elie Roumet, Ludovic Matar, Raksmy Nop, Geoffrey Daniel, Clément Gauchy, et al.. Boiling phenomena analysis using machine learning. ICMF 2025 - 12th International Conference on Multiphase flow, May 2025, Toulouse, France. Ludovic Matar, Clément Moulin-Frier, Eric Jenn, Christophe Marabotto. Application of curiosity driven exploration methods for hardware interference identification.	
Hobbies	
Running	Symphonic Concerts Piano: I interpret romantic and baroque music.